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DUNMAN HIGH SCHOOL Preliminary Examination Year 6

H2 BIOLOGY

9744/01

Paper 1 Multiple Choice Questions

23 September 2022

Additional Materials: Multiple Choice Answer Sheet

1 hour

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your centre number, index number, name and class at the top of this page.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

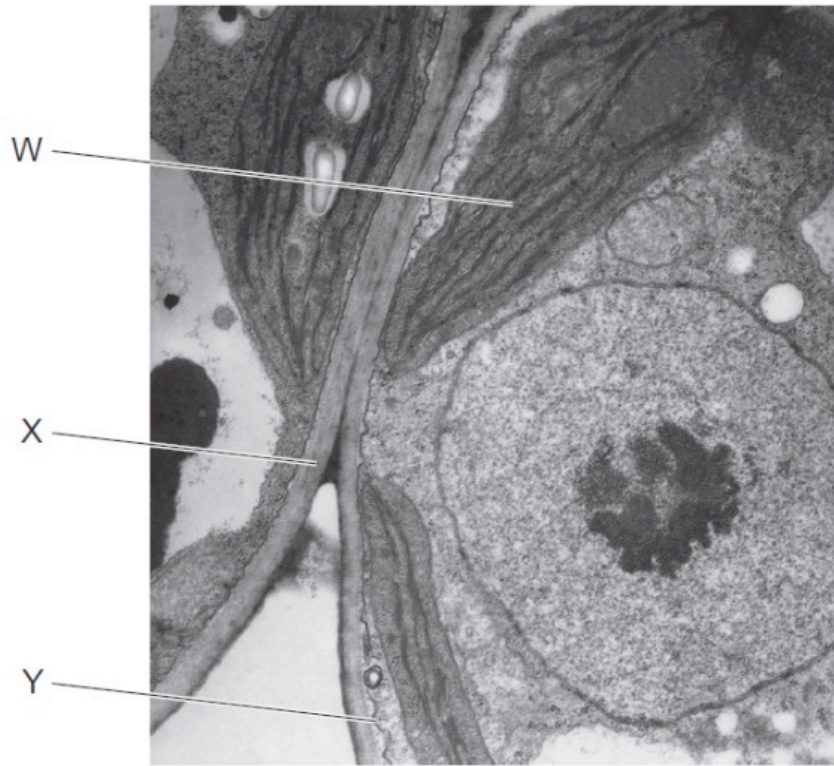
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

This document consists of **25** printed pages and **1** blank page.

1 The electron micrograph shows part of two cells.



Which of the labelled features enable these two cells to be identified as eukaryotic?

- A** W
- B** W and X
- C** W, X and Y
- D** None of the above

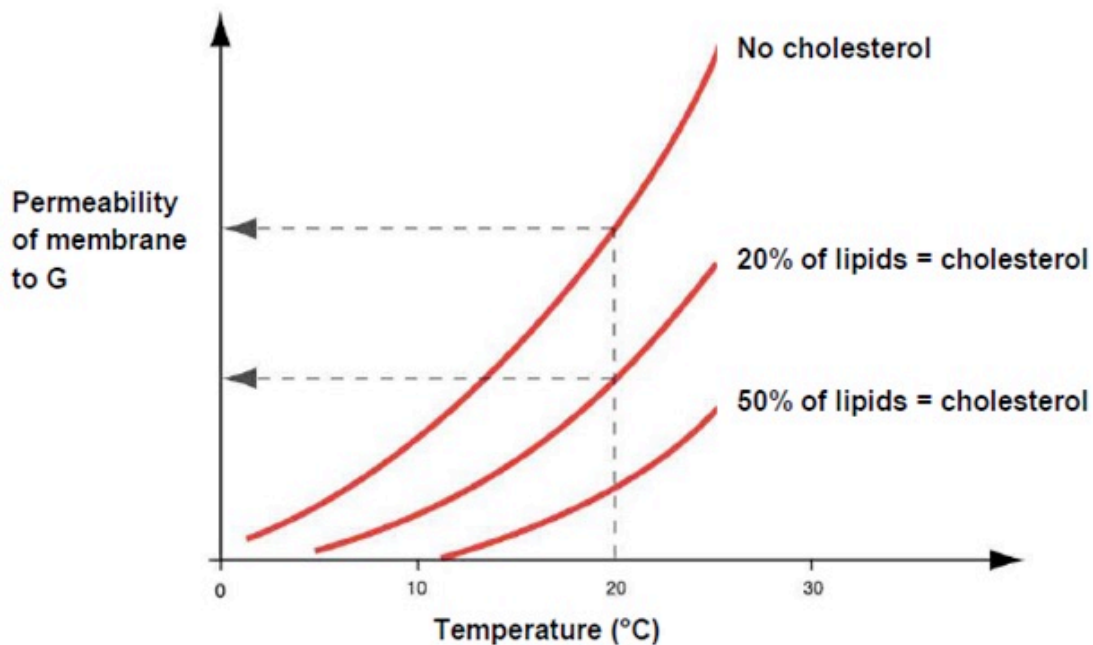
- 2 A sample of yeast cells was grown in a culture. Radioactive amino acids were added to the solution in which they were being grown. At various times, samples of the cells were taken and the amount of radioactivity in different organelles was measured. The results are shown in the table.

Time after radioactive amino acids were added to the solution/ minutes	Amount of radioactivity present/arbitrary units		
	P	Q	R
1	21	120	6
20	42	68	6
40	86	39	8
60	76	28	15
90	50	27	28
120	38	26	56

Which best describes the identities of organelles P, Q and R?

	P	Q	R
A	Golgi apparatus	Rough endoplasmic reticulum	Secretory vesicles
B	Rough endoplasmic reticulum	Secretory vesicles	Golgi apparatus
C	Rough endoplasmic reticulum	Golgi apparatus	Nucleus
D	Golgi apparatus	Secretory vesicles	Transport vesicles

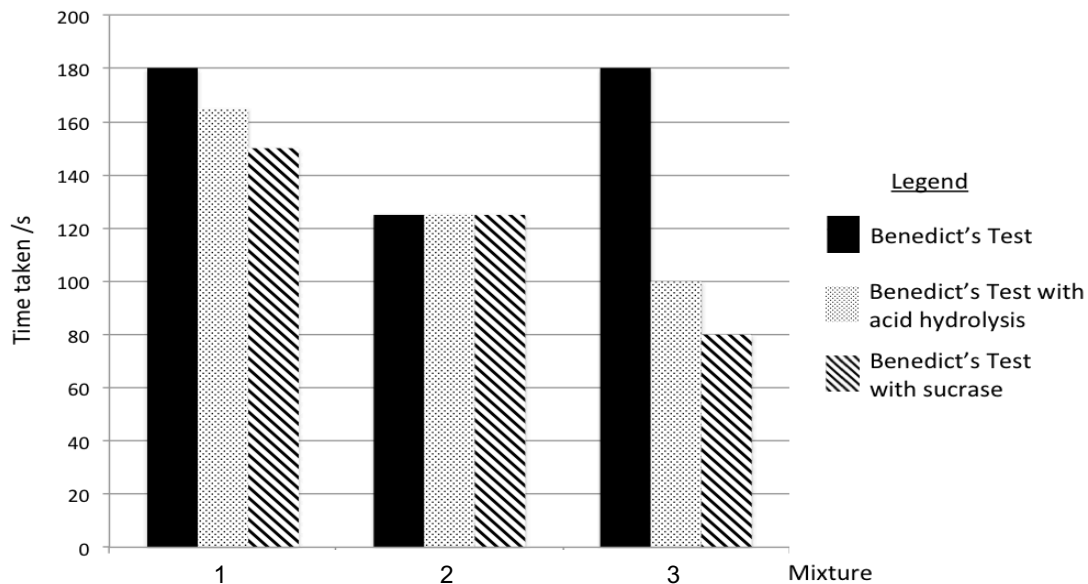
- 3 The graph shows the permeability of three different membranes to chemical **G** at different temperatures. These three membranes differ in the amount of cholesterol present in the phospholipid bilayers.



Which of the following is a possible explanation for the observed data?

- A** Increase in temperature increases the permeability of the membrane to **G** as cholesterol increases the fluidity of the phospholipids.
- B** At 20°C, an increase in the proportion of cholesterol in the membrane increases the permeability of the membrane to **G** as both cholesterol and **G** are non-polar.
- C** Increase in the proportion of cholesterol decreases the permeability of the membrane to **G** as cholesterol decreases the fluidity of the phospholipids.
- D** With an increase in the proportion of cholesterol in the membrane, a lower temperature is required to achieve the same level of permeability for **G** as **G** will gain a higher kinetic energy to penetrate the membrane.

- 4 Three mixtures (1, 2 and 3) containing different concentrations of glucose and sucrose were obtained. In order to determine the relative concentration of each substance in the mixture, the mixtures were subjected to different conditions before carrying out Benedict's Test. The time taken for brick-red precipitate to form was recorded in the graphs below.



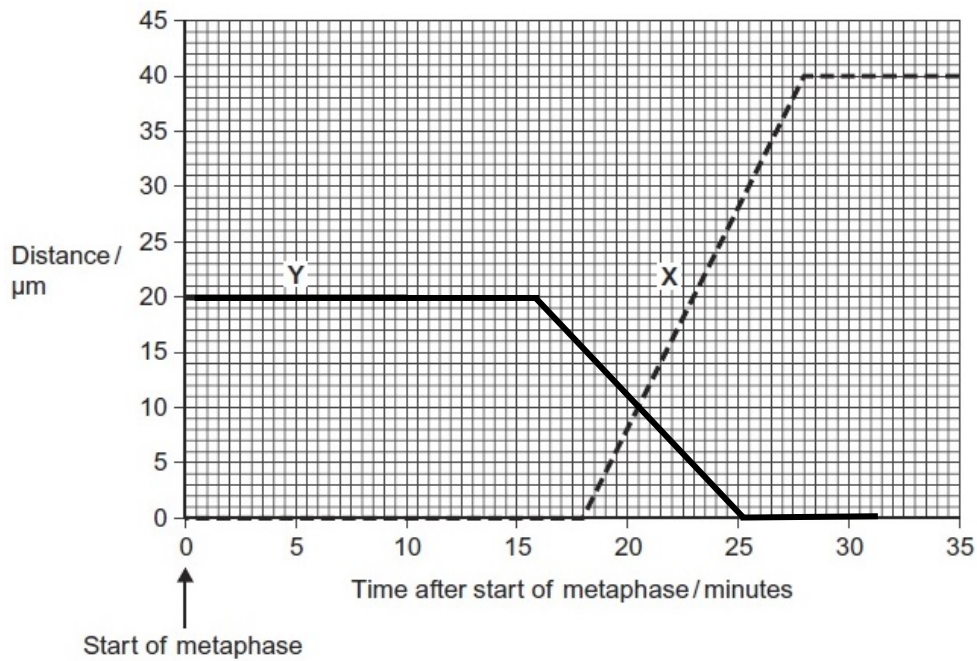
Which statement can be concluded from these graphs?

- A Mixture 2 contains equal concentrations of glucose and sucrose.
 - B Mixture 1 has a higher concentration of glucose than Mixture 2.
 - C Mixture 3 has a higher concentration of sucrose than Mixture 1.
 - D Mixture 3 has a lower concentration of sucrose than Mixture 1.
- 5 Each statement describes a similarity or difference between mitosis and meiosis.

Which statement is false?

- A Mitosis always produces diploid cells while meiosis always produces haploid cells.
- B Mitosis always results in two daughter cells while meiosis always results in four daughter cells.
- C Both mitosis and meiosis can occur in the absence of centrioles.
- D Both mitosis and meiosis involve the separation of sister chromatids.

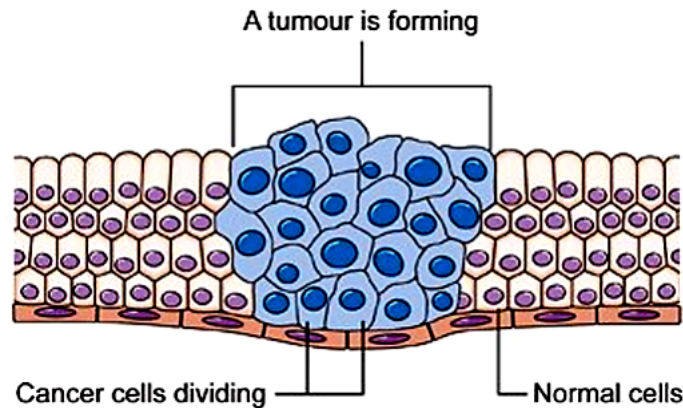
- 6 The graph shows information about the movement of chromatids in a cell that has just started metaphase of mitosis.



Which row matches the stages happening at X and Y indicated?

	Duration of metaphase / min	Increasing distance at X is due to	Measurement of X	Measurement of Y
A	35	segregation of sister chromatids	distance between each chromatid and the pole	distance between chromatids
B	18	segregation of sister chromatids	distance between chromatids	distance between each chromatid and the pole
C	35	chromosomes align and segregate	distance between chromatids	distance between each chromatid and the pole
D	18	chromosomes align and segregate	distance between each chromatid and the pole	distance between chromatids

- 7 The diagram shows cancerous cells dividing in between normal cells.



Which statements can be concluded from the diagram?

- 1 Cancerous cells can undergo self-renewal due to presence of telomerase.
- 2 Cancerous cells invade normal cells by undergoing excessive mitosis.
- 3 A tumour is a localised mass of cells.
- 4 A tumour is formed after undergoing metastasis and angiogenesis.

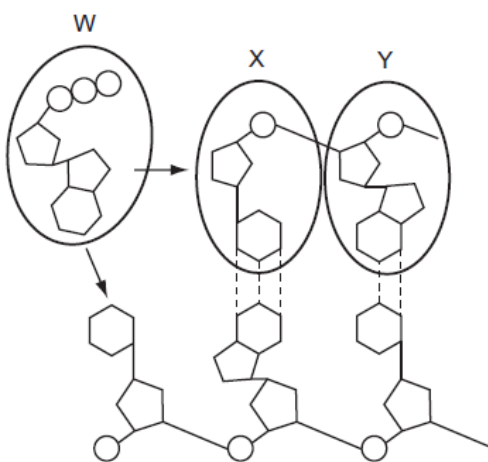
- A** 1 and 2
B 2 and 3
C 1, 2 and 4
D All of the above

- 8 Which statements regarding the structure of RNA are correct?

- 1 mRNA is a single-stranded, straight polynucleotide.
- 2 tRNA is a double-stranded polynucleotide that has a three-dimensional L-shaped structure.
- 3 rRNA is made up of nucleotides with the presence -OH group on carbon-2 of pentose sugar.

- A** 2
B 3
C 2 and 3
D 1, 2 and 3

- 9 The diagram shows the synthesis of a polynucleotide using a template strand. Molecule W is a nucleotide triphosphate.



Which statements are correct?

- 1 The base in W could be the purine, adenine.
 - 2 The base in Y is the purine, guanine.
 - 3 The base in X is the pyrimidine, cytosine.
 - 4 The base in X could be the pyrimidine, uracil.
- A** 1, 3 and 4
- B** 2 and 3
- C** 1, 2 and 4
- D** 1 and 3

- 10** In 1958, an experiment was published by Meselson and Stahl investigating the way in which DNA replicates.

Escherichia coli bacteria were grown in a medium containing $^{15}\text{NH}_4\text{Cl}$. After very many generations, virtually all of the bacterial DNA contained ^{15}N and the DNA was described as 'heavy'.

The bacteria were then transferred to a medium containing $^{14}\text{NH}_4\text{Cl}$. The bacteria were allowed to reproduce for three generations in this medium. The bacterial DNA from each generation was extracted and the percentage of DNA molecules containing ^{14}N DNA was determined.

What is the percentage of DNA molecules containing ^{14}N in the third generation?

- A** 0%
 - B** 25%
 - C** 75%
 - D** 100%
- 11** Part of the sense strand (non-template) in DNA is shown below.

3' CATGCGAATT 5'

Which of the following is the sequence of mRNA?

- A** 5' GUACGCUUAA 3'
- B** 5' AAUUCGCAUG 3'
- C** 5' CAUGCGAAUU 3'
- D** 5' UUAAGCGUAC 3'

- 12 Part of the amino acid sequences in normal and sickle cell haemoglobin are shown.

Normal haemoglobin	Sickle cell haemoglobin
thr-pro-glu-glu	thr-pro-val-glu

Possible mRNA codons for these amino acids are:

Amino Acid	mRNA codons
Glu	GAA GAG
Thr	ACU ACC
Pro	CCU CCC
Val	GUA GUG

Which tRNA molecule is not involved in the formation of this part of the sickle cell haemoglobin?

A



B



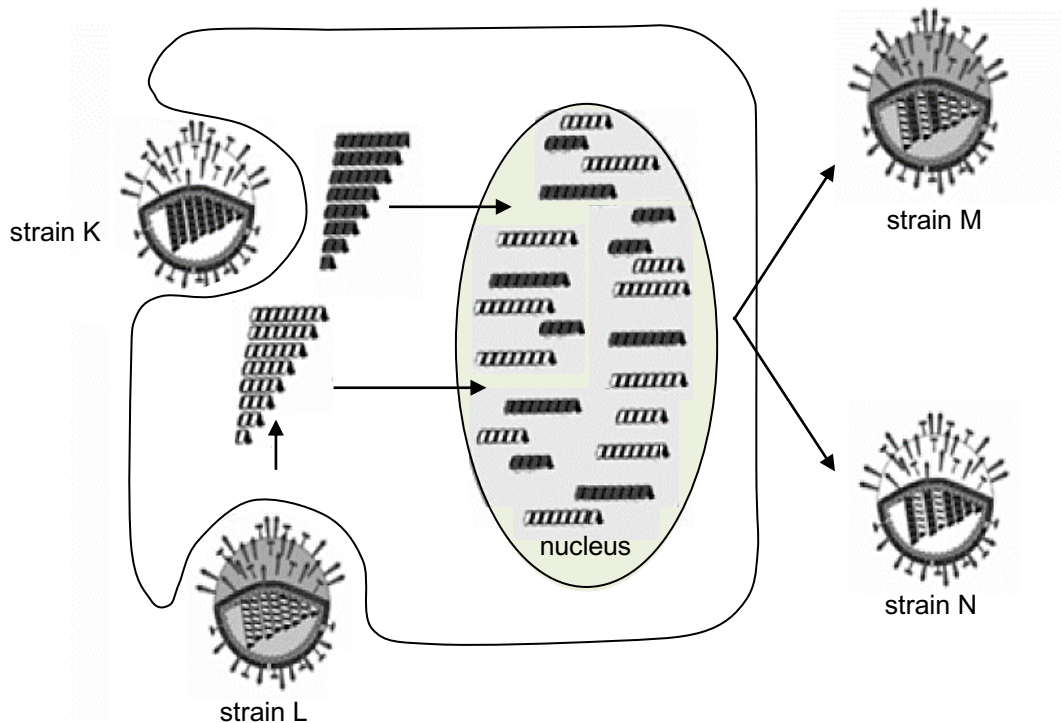
C



D



- 13 The diagram shows the emergence of new influenza strains, M and N, resulting in antigenic shift in influenza A.

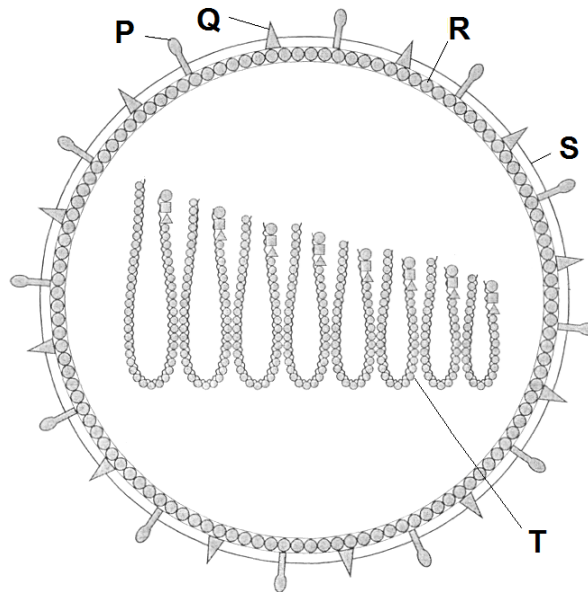


Which events lead to antigenic shift?

	Name of event	Description
1	gene mutation	gene mutations in the viral genomes of strains K and L occurred in the cell resulting in strains M and N
2	recombination of viral genome	recombination of strains K and L resulted in new strains M and N
3	reassortment of viral genome	reassortment of the viral genomes of strains K and L occurred only when both strains infect the same cell

- A** 2
B 3
C 2 and 3
D 1, 2 and 3

14 The diagram shows the structure of an influenza virus.



Which statements concerning the lettered components are correct?

- 1 Mutations that disrupt the function of **R** will result in the inability of the virus to initiate infection in the host cell.
- 2 **P** and **Q** are unlikely targets for vaccination because they undergo mutation constantly.
- 3 New influenza viruses acquire **S** from host cell during budding.
- 4 The host cell enzymes are not required to form the complementary RNA from **T**.

- A** 1 and 2
- B** 3 and 4
- C** 1, 2 and 3
- D** 2, 3 and 4

- 15** Binary fission allows bacteria to divide and gives rise to genetically identical daughter cells. Scientists are interested in uncovering the genetic mechanisms that regulate and drive binary fission, to design novel antibiotics that could specifically target and interfere with the process.

They found out that during binary fission, bacteria produce the protein FtsZ which assemble into a ring-like structure at the centre of the cell. This causes the cytoplasm to divide into two, before synthesis of a new wall begins.

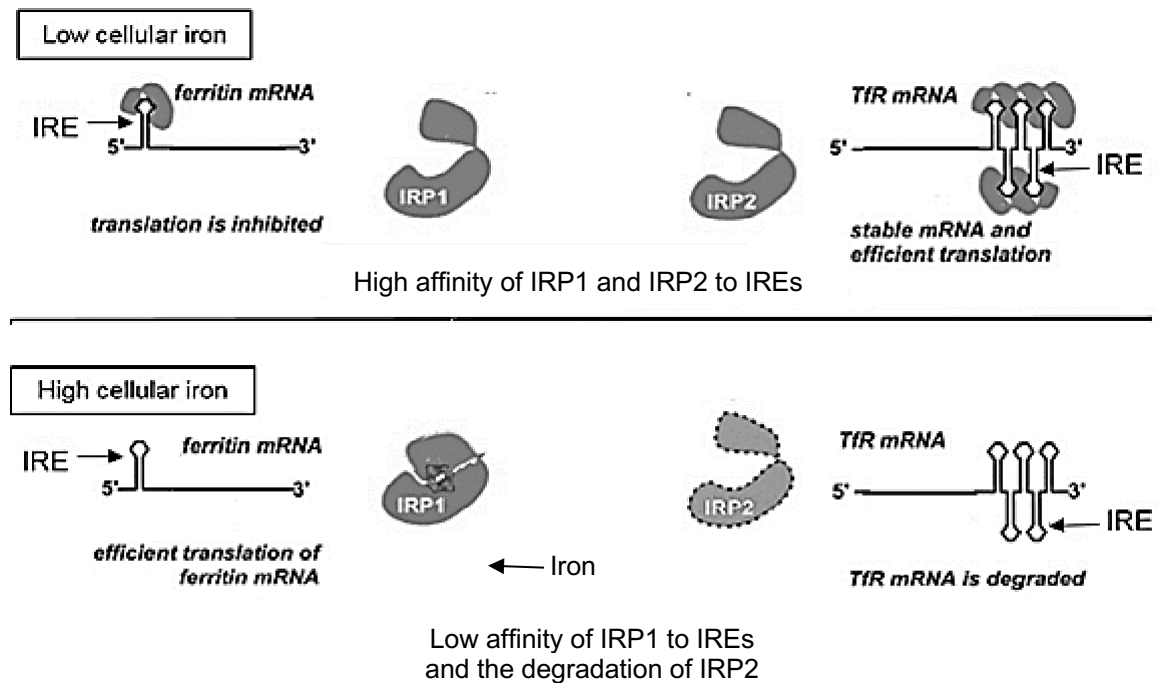
Which of the following statements can be deduced from the description above?

- 1 The genes that code for FtsZ can be found within a plasmid.
- 2 An antibiotic that inhibits the action of FtsZ will result in only one cell with more than one chromosome.
- 3 Mutations in genes coding for FtsZ will prevent the replication of DNA in binary fission.
- 4 The replicated chromosomes attach at the opposite ends of the parent bacterial cell before it divides into two daughter cells.

- A** 2
- B** 2 and 4
- C** 1 and 3
- D** 1, 2, 3 and 4

- 16** In mammalian tissues, iron enter cells via transferrin receptors (TfR) and are stored when they are bound by ferritin proteins. The synthesis of TfR and ferritin is regulated through translational controls by regulatory proteins IRP1 and IRP2, which bind to iron-responsive elements (IREs) found on mRNAs.

IREs are loops found at the 5' UTR of ferritin mRNA and 3' UTR of TfR mRNA respectively. The effects of different iron levels on IRP1 and IRP2 are shown in the figure.

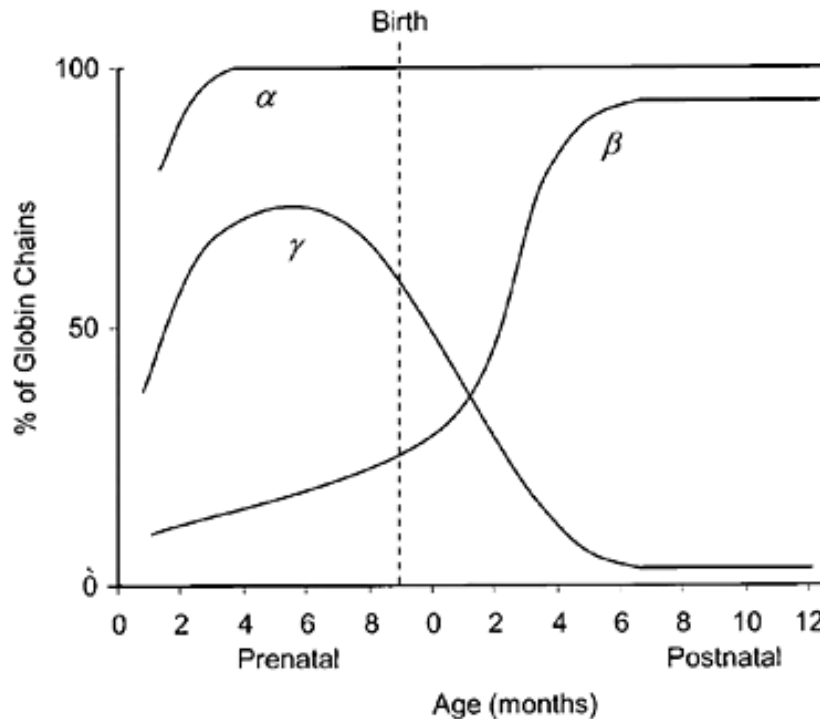


Which row correctly describes the translational control of ferritin mRNA and TfR mRNA?

	Cellular iron (Fe) levels	Ferritin levels	TfR levels
A	High	High	High
B	Low	Low	Low
C	High	High	Low
D	Low	High	Low

- 17 The globin gene family in humans consists of α , β and γ genes. These genes code for the globin chains that make up haemoglobin and are expressed at different levels during different developmental stages.

The graph shows the expression of the various globin chains during the prenatal (fetal) and postnatal (after birth) periods.



Which statement cannot account for the differences in the levels of expression of globin chains?

A	Methyl groups are added to regulatory sequences of γ -globin genes during the postnatal period, allowing for some proteins to bind.
B	A growth factor triggers the expression of a transcription factor that increases the rate of β -globin gene expression during the postnatal period.
C	Alternative splicing occurs in the mature mRNA of the α -globin and β -globin genes, resulting in differences in the rate of expression of globin chains during the prenatal period.
D	The shortening of poly(A) tail in the mRNA of globin genes reduces its stability, resulting in a decrease in the rate of expression of γ -globin chains during the postnatal period.

- 18** Two genes, Q and R, affect the size of the petals and the pigmentation of a flower.

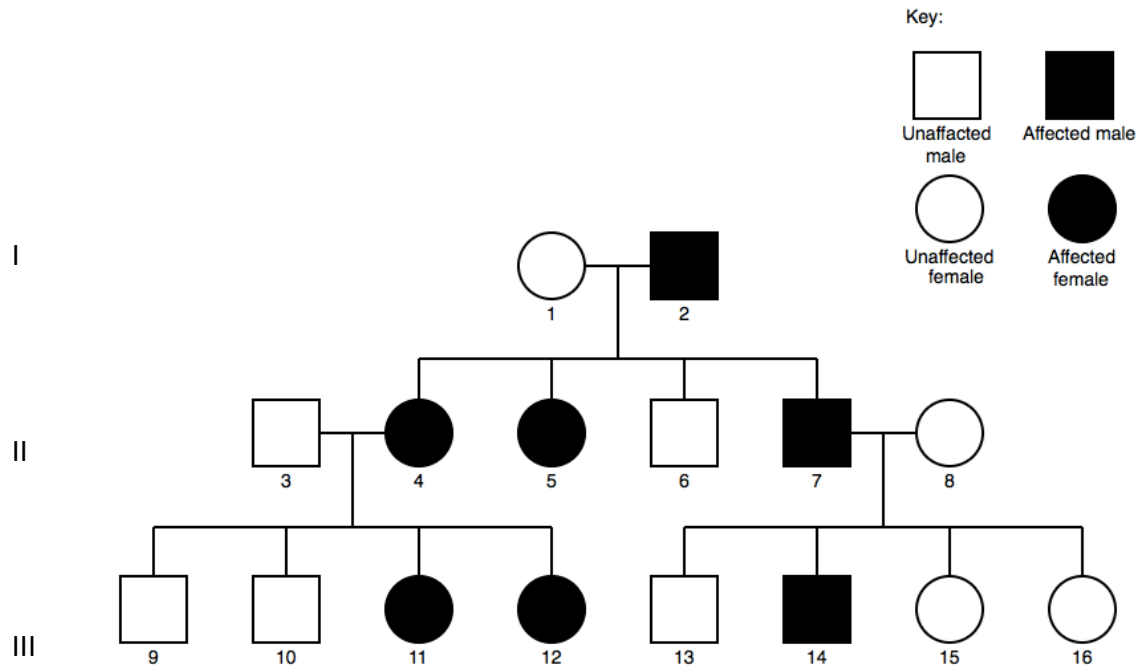
Gene Q has two alleles, Q^L and Q^A . The genotype Q^LQ^L produces large petals, Q^LQ^A produces small petals, and in Q^AQ^A , petals are absent.

Gene R has two alleles. R produces a red pigment and is dominant over the allele r that produces no pigment.

A plant that is heterozygous at both gene loci was selfed. How many different phenotypes will be observed in the next generation?

- A** 4
- B** 5
- C** 9
- D** 12

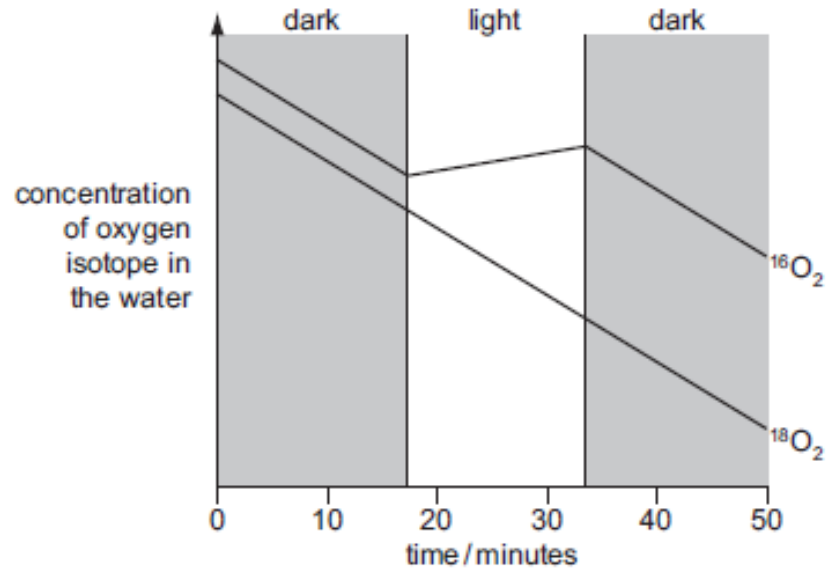
19 The figure shows a pedigree for the inheritance of polycystic kidney disease.



If the letters R and r denote dominant and recessive alleles respectively, what is the mode of inheritance for this disease and the most probable genotype of individual 11?

	Mode of inheritance	Genotype of individual 11
A	Autosomal dominant	RR
B	Autosomal recessive	rr
C	X-linked dominant	$X^R X^r$
D	X-linked recessive	$X^r X^r$

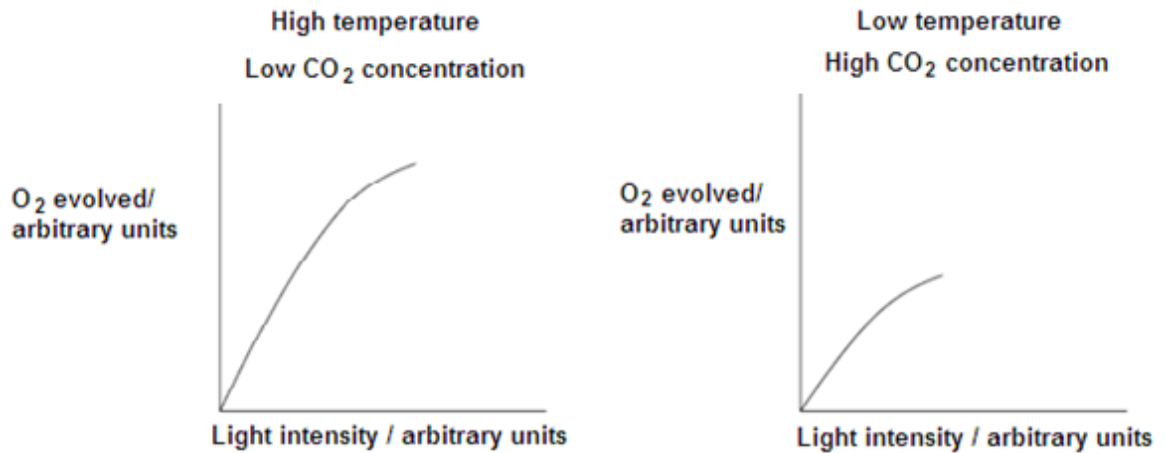
- 20** The common isotope of oxygen is ^{16}O . Air containing $^{16}\text{O}_2$ and $^{18}\text{O}_2$ was bubbled through a suspension of algae for a limited period. After this, the concentration of these two isotopes of oxygen in the water was monitored for the next 50 minutes whilst the algae were subjected to periods of dark and light. The results are shown in the diagram.



Which is the best explanation for these results?

- A** Both isotopes of oxygen are used by the algae in the dark in respiration, but in the light, oxygen is produced from water in photorespiration.
- B** The algae can distinguish chemically between the two isotopes.
- C** The algae produce oxygen from the water used in photosynthesis, but only in the light.
- D** The two isotopes have different rates of diffusion.

- 21 Students investigated the rate of photosynthesis by measuring the volume of oxygen evolved from an aquatic plant. The results of two experiments that they set up are shown below.

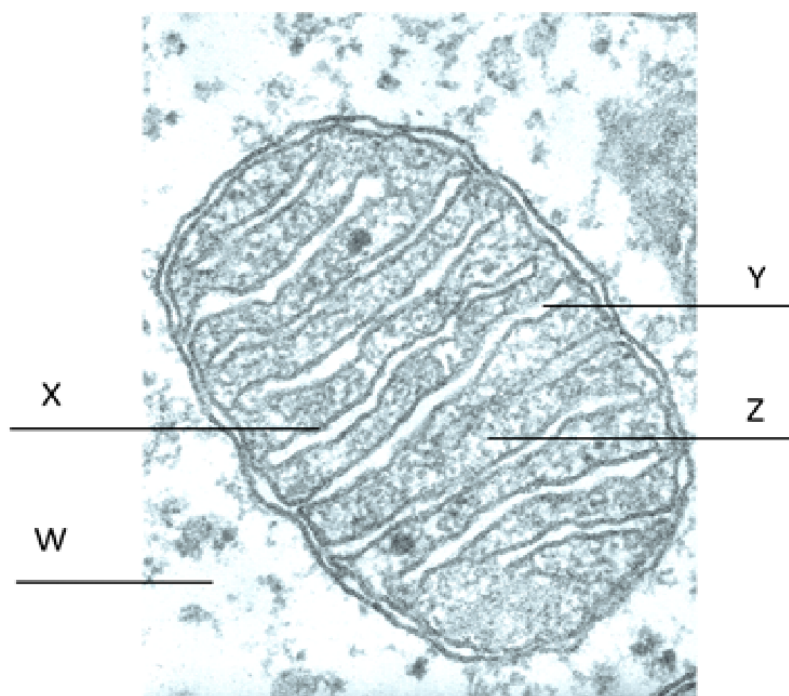


Which conclusion can be drawn from this data?

- A Temperature does not affect the rate of photosynthesis.
 - B High concentrations of CO₂ reduce the rate of photosynthesis.
 - C Temperature and CO₂ concentration are both limiting factors.
 - D The greater the light intensity, the greater the rate of photosynthesis.
- 22 After vigorous exercise, changes occur in the muscle tissues. Comparing with resting muscles tissues, what will the changes be?

	ATP	lactate	pH
A	decreased	increased	decreased
B	increased	no change	increased
C	decreased	decreased	increased
D	increased	decreased	no change

23 The figure shows an electron micrograph of an organelle.



Which row correctly matches the respective processes with the corresponding structures?

	Formation of pyruvate	Oxidative phosphorylation	Direction of diffusion of H^+ ions	Formation of reduced co-enzymes
A	W	X	$Z \rightarrow Y$	Z
B	Z	Y	$Z \rightarrow Y$	W and Z
C	Z	Y	$Y \rightarrow Z$	W
D	W	X	$Y \rightarrow Z$	W and Z

- 24** The hormone insulin binds to the tyrosine kinase receptors and initiates various signal transduction pathways to generate cellular responses. Which of the following shows the correct sequence of events, following the binding of insulin to the receptor?

- 1 phosphorylation of tyrosine residues
- 2 signal amplification
- 3 dimerisation of tyrosine kinase receptor
- 4 signal transduction
- 5 activation of transcription factors

- A** 1 → 3 → 2 → 4 → 5
- B** 3 → 4 → 1 → 2 → 5
- C** 1 → 3 → 5 → 4 → 2
- D** 3 → 1 → 4 → 2 → 5

- 25** Nerve cells in the brain communicate by chemical molecules known as neurotransmitters. The following events, in the order shown, describe how the binding of a neurotransmitter to a nerve cell will elicit a cellular response.

- 1 Neurotransmitter binds to a G-protein linked receptor (GPLR) at the cell surface
- ↓
- 2 A change in conformation of the GPLR allows the binding of G proteins to its intracellular domain
- ↓
- 3 G protein releases a GDP molecule in exchange for GTP
- ↓
- 4 The alpha subunit of the G protein dissociates from the other subunits
- ↓
- 5 The alpha subunit of the G protein activates adenylyl cyclase, which in turn converts ATP into cyclic AMP (cAMP)
- ↓
- 6 cAMP activates protein kinase A (PKA), which in turn phosphorylates transmembrane protein channels
- ↓
- 7 Phosphorylated protein channels open, allowing the facilitated diffusion of specific ions into the nerve cell

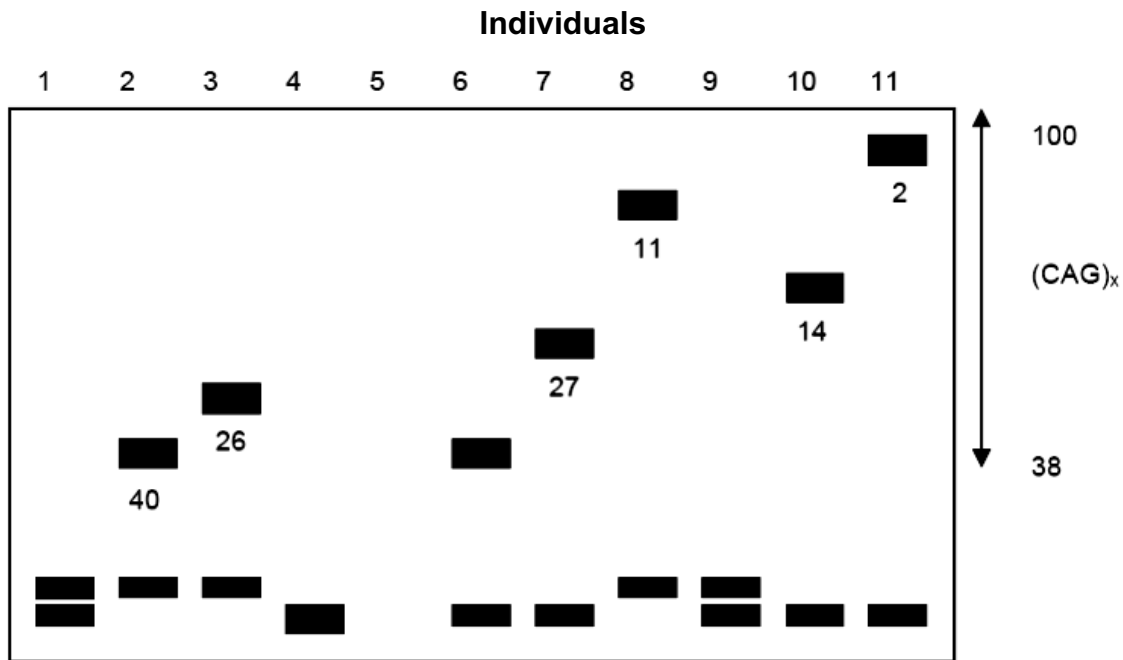
Which events allow for the amplification of signal to occur?

- A** 2, 5 and 6
- B** 1, 2, 5 and 7
- C** 3, 4, 5, 6 and 7
- D** 1, 4 and 6

- 26** The data shows the results of electrophoresis of PCR fragments amplified using primers for the site that has been shown to be altered in Huntington's disease.

The inherited mutation in the Huntington's disease gene abnormally repeats the nucleotide sequence CAG from 36 up to more than 120 times of that. The male parent, shown as individual 2, had the onset of Huntington's disease when he was 40 years old.

Six of his children (individuals 3, 5, 7, 8, 10, 11) suffer from Huntington's disease, and the age at which the symptoms first began is shown by the number below the band from the PCR fragment.



What is the likely outcome for the normal individuals 4, 6, and 9?

- A** Individuals 4 and 9 do not have the trait, and will not get Huntington's disease, but individual 6 is likely to start the disease when he reaches his father's age of 40.
- B** Individuals 4, 6, and 9 have not inherited the defect causing Huntington's disease.
- C** Individuals 4, 6, and 9 will still develop Huntington's disease at some point in their lives, since the disease is inherited as a dominant trait.
- D** Two of the three will develop the disease, since it is inherited as a dominant trait, but the data does not allow you to predict which two.

- 27** The DNA for a heavy chain in a B-cell making IgG₄ antibody specific for retroviruses has the following structure: 5'–V₁₄D₃J₂C_{γ4}–C_ε–C_{α1}–C_{α2}–3'. Knowing that IgM is the first antibody formed, how many individual rearrangements were required to go from the embryonic DNA to this B-cell DNA?

A 1
B 2
C 3
D 4

- 28** A monoclonal antibody specific for a virus was produced by identical plasma cells.

This antibody was treated with an enzyme to break the bonds between the variable and constant regions.

The separated variable and constant regions were then added to cells infected with the virus.

With regard to the experiment carried out, which statements are correct?

- 1 The constant regions would bind to different parts of the virus antigens.
- 2 The variable regions would all bind to the same part of the virus antigens.
- 3 The separated constant and variable regions cannot bind to the virus antigens.
- 4 Phagocytes will engulf the viruses.

A 2
B 3
C 1 and 2
D 2 and 4

- 29** Which statement does not explain why the population is the smallest unit that can evolve?
- A** Natural selection involves competition between individuals in a population.
 - B** Evolution occurs when allele frequency in a population changes due to selection or chance events like infections with lethal diseases.
 - C** Differential reproductive success is observed at the population level due to the phenotypic variations in the population.
 - D** Evolution involves the introduction of advantageous mutations into the gene pool of a population as a result of a selective pressure.
- 30** Which effect of temperature increase on arctic ecosystems, will increase carbon dioxide in the atmosphere?
- A** Greater production of plants due to warmer temperatures and changing vegetation.
 - B** Melting ice from glaciers and icebergs will cause sea levels to rise.
 - C** Less ice and snow will cause incoming radiation to be absorbed more readily.
 - D** Greater decomposition of organic matter currently stored in permafrost.

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